

Project Initialization and Planning Phase

Date	01 July 2026
Team ID	SWTID-2026-8998
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to automate the credit card approval process using machine learning techniques, improving the speed, accuracy, and consistency of approval decisions. The system analyzes an applicant's financial and demographic information to predict whether the credit card application should be approved or rejected. By reducing manual verification and human error, the solution enables banks to make reliable, real-time credit approval decisions while enhancing customer experience.

Project Overview	
Objective	The primary objective is to develop an AI-powered credit card approval prediction system that accurately predicts application approval using historical customer data and machine learning algorithms.
Scope	The project focuses on collecting applicant information, preprocessing data, training multiple machine learning models, selecting the best-performing model, and deploying it through a Flask web application with IBM Watson Machine Learning for real-time predictions.
Problem Statement	
Description	Traditional credit card approval systems rely heavily on manual verification, making the approval process time-consuming, inconsistent, and prone to human error. Applicants often experience delays and uncertainty regarding their eligibility.
Impact	Automating the approval process using machine learning reduces processing time, improves prediction accuracy, minimizes operational costs, and enhances customer satisfaction by providing faster and more transparent approval decisions.
Proposed Solution	
Approach	Develop a machine learning-based credit card approval prediction system using applicant financial and demographic information. The system trains multiple classification models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, and deploys the best-performing model using Flask and IBM Watson Machine Learning.

Key Features	- AI-based Credit Card Approval Prediction - Multiple Machine Learning Model Comparison - Applicant Eligibility Assessment
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, pickle
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Dataset	Credit Card Approval Dataset (CSV), Applicant Financial & Demographic Data